ABHAY SINGH

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LinkedIn • GitHub • Portfolio

PROFESSIONAL SUMMARY

Data Engineer with **3+ years** of production experience designing and operating large-scale data platforms on Azure and Databricks. Specialized in Apache Spark/PySpark, Delta Lake, and Medallion Architecture (Bronze / Silver / Gold), with deep focus on **platform automation**, CI/CD-driven pipelines, event-driven data processing, and cloud cost optimization. Most recently, solo-designed and deployed **ARIA** — a production agentic AI system that reduced L2 pipeline failure investigation from 30–60 minutes to 2–5 minutes, built on a custom LLM agent loop (Azure OpenAI / Databricks-hosted Claude) orchestrating 23 tools across ADF, Databricks, ServiceNow, and Azure DevOps. Proven track record of cutting release cycle times, eliminating 16+ engineer-hours of manual effort weekly, and delivering ~$848K in infrastructure savings. **Databricks Certified Data Engineer Professional & Associate.**

TECHNICAL SKILLS

Languages	Python, Scala, SQL
Processing & Streaming	Apache Spark (Core, SQL, Structured Streaming), PySpark, Apache Kafka, Event-Driven Architecture
Storage & Lakehouse	Delta Lake, Medallion Architecture (Bronze / Silver / Gold), ADLS Gen2, Azure SQL, MinIO
Cloud & DevOps	Azure Data Factory, Azure Databricks, Databricks Notebooks, Azure DevOps REST API, Docker, CI/CD Pipelines
Orchestration	Apache Airflow, ADF Pipelines
AI & Agents	LLM Agent Design (OpenAI Function-Calling), Azure OpenAI, Databricks-hosted LLMs (Claude Sonnet), Agentic Tool Orchestration, Streamlit
Data Quality	Schema Evolution, Data Reconciliation, Idempotent Pipelines
Monitoring	Prometheus, Grafana, Azure Logic Apps, Custom Pipeline Logging & Alerting

WORK EXPERIENCE

Data Engineer — Xebia IT Architects

Feb 2023 – Present | Gurugram, India | Client: Sephora (Retail Data Platform)

Dual-team role — one of few engineers simultaneously across: **L2 Data Platform Support** (~20 members) covering production reliability, SLA-bound incident resolution, and data discrepancy triage; and the **Agile Data Engineering** team (~15 members) for new development, enhancements, and complex escalations. Senior member: oversees peers and mentors newcomers. *Scale: ~1,500 active Databricks jobs/ADF pipelines, 100s of GBs/day across a PB-scale lakehouse.*

ARIA — Automated Root-cause Intelligence Agent

Python • Azure OpenAI (GPT-4.1-mini) • Databricks-hosted Claude Sonnet 4.6 • Databricks Apps • Streamlit • Delta Lake

- Solo-designed and deployed a production agentic AI triage system that replaces manual L2 pipeline failure investigation — reducing diagnosis time from **30–60 minutes to 2–5 minutes** per incident across the Sephora data platform (~1,500 active jobs/pipelines).
- Built a custom multi-step LLM agent loop (OpenAI function-calling, up to 15 iterations) orchestrating **23 tools** across ServiceNow, ADF, Databricks, Azure DevOps, and Spark SQL; LLM provider is fully config-switchable between Azure OpenAI and Databricks-hosted Claude Sonnet with no code changes required; Delta-backed preservance cache (24h TTL) skips re-triaging known failures.
- Supports two entry points: **Pipeline mode** (ADF pipeline name → full agent loop fetching run details, stack traces, config files, and Git diffs) and **INC mode** (ServiceNow INC → LLM extracts pipeline → full triage → automatic SN write-back); scheduled **auto-scan job** runs every 30 min to proactively triage new incidents.
- Performs live ADO commit log analysis and file diff inspection to detect **CODE_CHANGE** failures; runs Spark SQL queries against prod tables for data quality checks — producing a 7-class failure taxonomy with HIGH/MEDIUM/LOW confidence and a full evidence chain per result.

ADF Multi-Environment Release Pipeline Automation

Python • Azure DevOps Pipelines • Azure Data Factory • Databricks • CI/CD • Git

- Sole architect and builder of a release engineering overhaul: centralised all ADF pipeline definitions in a single Azure

DevOps repo with environment-specific default branches (dev/qa/master), replacing manual copy-paste releases across environments.
- Built a parameterised Azure DevOps pipeline accepting a release ID and change list (pipelines, triggers, linked services, global parameters) — captures current QA state, creates a versioned release branch, and auto-raises a PR. Engineers just approve and merge; every release is fully auditable via Git history.

Sales Promotion Ingestion Platform

Scala • Apache Spark • Delta Lake • SFTP • Medallion Architecture (Bronze / Silver / Gold)

- Led redesign of a legacy JDBC ingestion process; replaced with a distributed Scala + Spark framework over SFTP-driven Medallion architecture, standardizing data contracts and improving fault tolerance.
- Built Delta Lake reconciliation workflows enforcing schema consistency, primary key alignment, and business rule parity — validated **50M+ historical records** with zero duplication and full referential integrity.
- Developed scalable backfill and comparison jobs supporting incremental loads and controlled schema evolution across environments.

ADF Pipeline Update Automation & Platform Tooling *github.com/ADFUpdateAutomator*

Python • Azure DevOps REST API • Azure Data Factory • Databricks • CI/CD

- Owned and redesigned manual ADF release workflows; automated branch creation, pipeline JSON modification, validation, and PR generation end-to-end via Python and the Azure DevOps REST API.
- Built JAR-version comparison logic that checks existing pipeline configs before writing — eliminating redundant updates and config drift, reducing release defects by **25%**.
- Enabled fully CI/CD-driven ADF releases with automated PR creation and direct URL surfacing inside Databricks notebooks, cutting update time by **90%** and overall release cycle by **30%**.
- Implemented conflict-safe branch management (reuse, recreate, push-conflict handling) — hardening the tool for concurrent team usage.

Legacy OPD Data Platform Modernization

PySpark • Databricks • Medallion Architecture • Delta Lake

- Migrated 30+ production tables and pipelines from a legacy source system to an updated schema, redesigning PySpark transformation logic across Bronze / Silver / Gold layers while maintaining full output parity for all downstream consumers.
- Ran cross-system validation across all 30+ affected tables before cutover — covering schema contracts, business rule parity, and referential integrity — while eliminating tightly coupled joins, improving data mart query performance by **25%**.

Additional Engineering Impact

- Built a centralised pipeline observability system — a shared Delta table capturing run statuses across all jobs, paired with a reusable Azure Logic App dispatching custom alert emails with attachments to configurable recipients based on SFTP file health, blob availability, and pipeline state.
- Optimized Spark and Databricks ETL pipelines via partition tuning, broadcast joins, and caching — **25%** throughput gain and measurable compute cost reduction.
- Reduced manual operational effort by **16+ engineer-hours per week** through targeted automation utilities, contributing to ∼**$848K** in annual cloud infrastructure savings.

PERSONAL PROJECTS

Real-Time Flight Monitoring System *github.com/real-time-flight-monitoring*

PySpark • Spark Structured Streaming • Apache Airflow • Delta Lake • OpenSky Network API

- Built an end-to-end real-time and batch hybrid data platform ingesting live global flight telemetry from the OpenSky REST API into a Bronze Delta Lake layer, orchestrated across three Airflow DAGs with process-liveness checks and auto-recovery.
- Implemented Spark Structured Streaming Silver-layer consumers with null/coordinate/ICAO validation, aircraft-ID + timestamp deduplication, and ingestion timestamping.
- Built Gold-layer batch enrichment every 5 minutes via Airflow — deriving flight phase flags, UTC timestamps, speed metrics, and country/hour aggregations.

Retail-360: Multi-Source Retail Analytics Platform *github.com/multi-source-retail-analytics*

Apache Kafka • PySpark • Airflow • Delta Lake • MinIO • Docker • Prometheus • Grafana

- Designed a production-style event-driven retail analytics platform ingesting transactional, inventory, and customer streams via Kafka into a Medallion lakehouse on MinIO (S3-compatible).
- Containerized the full stack (Kafka, Spark, Airflow, MinIO) using Docker Compose for reproducible local and cloud-parity deployments.
- Enforced schema contract validation between Bronze→Silver and Silver→Gold transitions before each layer promotion.
- Deployed Prometheus + Grafana observability for pipeline health, data freshness tracking, and job runtime alerting across all layers.

CERTIFICATIONS

- **Databricks Certified Data Engineer Professional** – Valid Jun 2026 – Jun 2028
- **Databricks Certified Data Engineer Associate** – Valid Nov 2023 – Jan 2028
- **Databricks Lakehouse Fundamentals** – Valid Jan 2025 – Jan 2026

AWARDS & RECOGNITION

- Galaxy Best Project Team (Taurus) Award – Xebia IT Architects
- GEM Award (×4) – Xebia IT Architects

EDUCATION

B.Tech. Computer Science — Specialization in AI & ML

Sushant University, Gurgaon, India Jul 2019 – Jun 2023